Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Bengaluru, India

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Summary

DevOps Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, shipping CI/CD, IaC, and multi-account governance for enterprise financial-services and technology customers (RWS, State Street, DTCC). Deep Terraform (modules, tftest, PyTest) plus GitHub Actions, CodePipeline, CodeDeploy, and blue-green delivery on ECS. Built Terraform with automated tests for about 10 AWS services at State Street, and now automate RWS's AWS foundation in Terraform Cloud, cutting per-account setup from 3+ days to under 4 hours. 11 upstream PRs merged across axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems, 107 contests.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Bengaluru, India

- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (~90%) by replacing a legacy account-vending pipeline with Terraform Cloud automation (OIDC) for a data-centre exit and a second landing zone: 4,000+ resources across 20+ workspaces, none failing.
  - Codified Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines with IAM Identity Center, AWS Backup, and SSM Patch Manager in every workload account; every change ran in a sandbox mirror first through the same least-privilege role.
  - Built self-service network onboarding on two Transit Gateway hubs: route table within 15 minutes, firewall access by tag in about a minute. Used Kiro and Claude Code for specs and design documentation.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a **10/10 CSAT**.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC with Amazon Q.
- **MLOps Pipeline (SME Program, Jun – Aug 2025):** Built end-to-end SageMaker pipeline for medical-image classification with blue-green endpoint deployment, auto-scaling, and drift detection – provisioned via Terraform + CodePipeline with automated rollback on canary failure. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Published a second AWS sample on Organizations governance (see Publications). Co-authored an AWS Prescriptive Guidance pattern on secure PR previews and approval-gated releases for Amplify Gen 2. Delivered 2 org-wide tech talks on Terraform testing and GitOps pipeline gating. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps – AWS DevOps Agent, Agentic AI, Serverless). MLOps SME Program Graduate. Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – enabling zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for cross-service latency visibility.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code-quality analysis into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Projects

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

skillcheck Jul 2026
TypeScript, Node.js, GitHub Actions, SARIF GitHub | Marketplace
◇ Conformance linter for Agent Skills: 36 rules with message-for-message parity to the skills-ref reference validator, SARIF 2.1.0 output and inline PR annotations. Ships as an npm CLI (284 downloads in the last month) and a GitHub Marketplace action.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Claude Cost Optimizer Mar 2026
Claude Code Plugin, MCP, TypeScript, React, Python GitHub | Live
◇ Claude Code plugin, MCP cost server, CLI grader and web calculator for Claude spend. Rebuilt every estimator on cache-aware billing, correcting a modelled 30-turn Opus session from $3.44 to $0.74, and replaced every vague savings range with a sourced or computed number. Golden-value tests pin the cost math in all three implementations.

Other Notable Projects: Bedrock Multi-Model MCP (26 text models, 156 of 156 benchmark calls), itr-agent (deterministic tax engine, 151 tests), DevOps-AWS-FARM (Docker + GitHub Actions OIDC deploy to ECR/ECS), GitScope (Chrome Web Store extension), Ledger Sync (self-hosted finance app, 26 pages).

Open Source Contributions (11 merged upstream)

- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

IaC & Config: Terraform (modules, tftest, PyTest, remote state, S3 native locking), HCL, CloudFormation, Ansible, Packer
CI/CD & Delivery: GitHub Actions, AWS CodePipeline, CodeBuild, CodeDeploy, Blue-Green & Canary Deployments, GitOps, Policy-as-Code
AWS Platform: Organizations, SCPs, Control Tower, Transit Gateway, Direct Connect, VPN, ECS Fargate, Lambda, EventBridge, API Gateway, CloudFront, KMS, IAM, Config, Security Hub
Observability & SRE: CloudWatch (Logs, Metrics, Alarms), X-Ray, SonarQube, Drift Detection, Auto-Scaling, Incident Runbooks
Languages & Runtime: Python, Bash, TypeScript, SQL, Docker, Linux
Compliance & Governance: CCMv4, Least-Privilege IAM, Preventative Controls, Tag Policies, Cost & Usage Reporting